The Regenerative Method

A field guide in ten chapters, from the family farm at Viking Farms PH, Calamba, Laguna. By Kathy Hingan, Daryl Urbanski, and Willy Hingan, the farmer.

Ten chapters, one per module of the method, in the order a transition actually happens: the case for changing, then soil, then the practices, then the first season, then the economics and the measurement that keeps it honest. Each chapter opens with the short answer, then the depth.

Everything in this guide was farmed before it was written, on the same land the family has kept since 1940 while the land around it turned industrial. The straw that used to burn is the mulch; the munggo on the paddies' edges is the cover layer; the notebook in the farmer's handwriting is the record the method is taught from.

The tropical calendar runs through every chapter: wet and dry seasons, typhoon resilience, rice paddies, and the timing that imported temperate advice gets wrong.

Contents

Chapter 1. Why Regenerative, Why Now

Chapter 2. Know Your Soil

Chapter 3. Compost and Biofertilizers

Chapter 4. Stop Tilling

Chapter 5. Cover the Soil

Chapter 6. Pests Without Poisons

Chapter 7. Water and Seasons

Chapter 8. The First-Season Conversion Plan

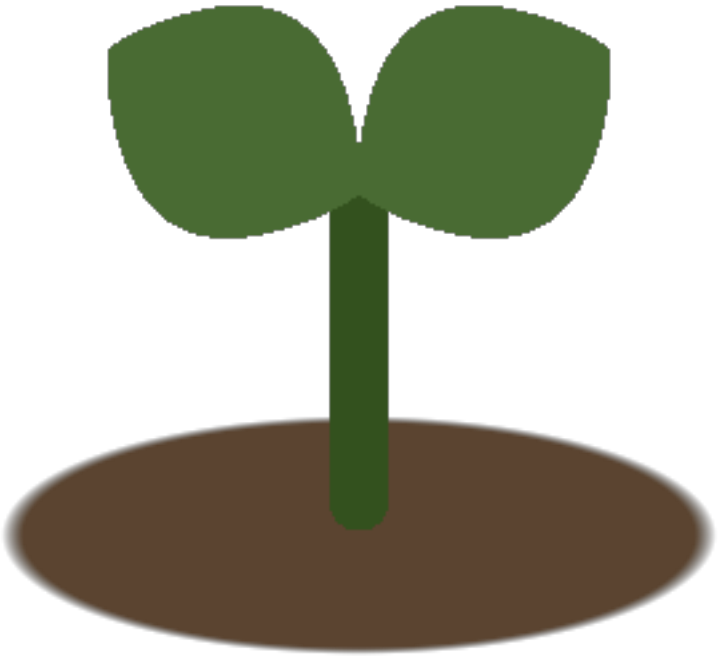
Chapter 9. Making It Pay

Chapter 10. Measure and Keep Going

Where to go from here

Chapter 1. Why Regenerative, Why Now

What regenerative agriculture actually means: the definition, the five practices, and how it differs from organic, sustainable, and permaculture.



Short answer. Regenerative agriculture is farming that rebuilds soil health instead of depleting it. It rests on five practices: living roots, covered soil, minimal disturbance, crop diversity, and feeding soil biology. It differs from organic by what it builds toward, not just what it avoids, and it is learned season by season.

A working definition, from a family that farms.

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The definition

Regenerative agriculture is a way of farming that rebuilds soil health instead of depleting it, by keeping living roots in the soil, keeping the soil covered, disturbing it as little as possible, growing a diversity of crops, and feeding the soil biology rather than only feeding the plant.

That is the working definition this whole site is built on. Notice what it does not say: it does not say regenerative agriculture is a product you buy, a certificate you earn, or a rulebook you follow. It is a direction of travel. A farm is regenerative to the degree that its soil, its biology, and its profitability are all improving year over year, and conventional to the degree that they are not.

This matters because the word is young. Wikifarmer's 2025 agrifood trends report notes that regenerative agriculture is "yet to be universally defined and standardized," which means no certification body owns the term and no single definition has won. That is not a weakness in the word. It is the reason a farmer can still decide for themselves what the standard should be, on their own land, and why we wrote ours down.

The five practices, in plain language

Every working definition of regenerative agriculture converges on a small set of practices. Ours has five, and each one is something a farmer can start this season:

Keep living roots in the soil. Roots feed the biology underground. Bare soil between crops is hungry soil. A cover crop or a cash crop, something growing, all year.

Keep the soil covered. Mulch, residue, a living cover. Covered soil holds water, resists weeds, and does not bake or wash away in the tropical sun and rain.

Disturb the soil as little as possible. Every pass of a tiller breaks the fungal networks and opens the soil to erosion. Less disturbance, more structure.

Grow a diversity of crops. Monoculture feeds one pest and one market. Diversity spreads both risks. On our farm that is the seven-layer planting: canopy trees, vines, cash crops, cover crops, roots, groundcover, and animals in the same system.

Feed the soil biology, not just the plant. Compost, manure, residues from what the farm already produces. The plant is fed by the biology; feed the biology and the plant follows.

If a farm does those five things, it is regenerating, whatever anyone calls it. If it skips them, no label fixes it.

What each practice replaces on a real farm

The practices are easiest to understand by what they replace. On our farm in Laguna, each one took over a purchased input or a chore the old way demanded:

Living roots replaced bare fallow between cash crops, which used to mean buying herbicide to keep the weeds down on empty ground.

Covered soil replaced hand-weeding as the main weed strategy, because mulch outcompetes germination instead of fighting the plant.

Minimal disturbance replaced routine tilling, which had become a twice-a-season chore that cost fuel and broke the soil's structure every time.

Diversity replaced the single-cash-crop year, which concentrated the family's income into one harvest and one market price.

Feeding the biology replaced bagged fertilizer as the default, using rice straw, manure, and residues the farm used to burn or haul away.

None of these five transitions cost money to start. Each one traded a recurring purchase for attention and timing, and that trade is the economics of the whole method.

How it differs from organic, sustainable, and permaculture

People arriving at this page are usually asking one of three comparison questions, and the differences are real enough to answer plainly:

Organic

Organic is a list of things a farm stops doing: no synthetic fertilizer, no synthetic pesticide. A farm can be certified organic and still till heavily, still leave soil bare between crops, and still lose topsoil every season. Organic is about what is absent from the field. Regenerative is about what is present and improving in it. Many regenerative farms are organic; organic does not require a farm to regenerate anything.

Sustainable

Sustainable means "able to continue." A farm can be sustainable and merely stop the bleeding, holding soil and profit steady without rebuilding anything. Regenerative aims higher: soil that is better next year than this one. In practice the words overlap enough that the market uses them loosely; the difference is the direction, maintenance versus improvement.

Permaculture

Permaculture is a design philosophy: permanent agriculture laid out as systems of zones, sectors, and guilds. It shares regenerative agriculture's respect for diversity and biology, and on a home garden scale the two are hard to tell apart. The practical difference is the working farm: permaculture design guides what to build; regenerative practice, as we teach it, is the season-by-season work of keeping a producing farm's soil alive while it pays.

Why the definition matters more in the Philippines

The imported advice fails here for one reason: timing. Most regenerative agriculture writing is temperate. It assumes a spring and a fall, a winter dormancy, and a growing calendar built for four seasons. The Philippines has a wet season and a dry one, typhoons instead of snow, and rice paddies that behave like nothing in the textbook. A definition is only useful if the practices underneath it survive translation.

That translation is the difference between this page and the hundreds of others answering the same question. The five practices above are not adapted to the tropics as an afterthought; they are taught here from a farm that runs them, in both seasons, in Laguna.

What regenerative agriculture is not

Equal and opposite to the definition, and just as useful:

It is not a product. No spray, bottle, or bag makes a farm regenerative. Companies sell inputs; the practices above are mostly about needing fewer of them.

It is not a certificate. No certifier owns the term yet. When one arrives, it will describe a farm that already regenerates; it will not make one.

It is not an all-or-nothing conversion. A farm transitions one field, one season at a time. The first cover crop on one paddies' edge is a start, not a failure to convert everything at once.

It is not anti-farmer. Conventional farming is what most farmers know, and it fed this family for generations before we changed. The transition is a business decision made under cost pressure, not a moral one, and the farmers who make it deserve the economics explained, not a lecture.

Where the word came from

The term entered farming through soil science and the grazing world, and it spread because it described something the older words did not: not merely keeping a farm going, but leaving the land better than it was found. That is also why it is worth defining carefully. A word that means "improving" can be claimed by anyone, and the only honest test of a claim is whether the soil under it is getting better, season after season, in a way a farmer can see.

Where to go next

The definition is module one of ten. What follows it on this site:

Know your soil : the living soil science in plain language, and the simple tests a farmer can run.

Compost and biofertilizers : fertility from what the farm already produces.

The first-season conversion plan : the step-by-step first year.

The Method : the full ten-module course.

Or take the Find Your Path quiz and get the module your farm should start with, based on what you grow and what is costing you most.

Frequently asked questions

What is regenerative agriculture in one sentence?

It is farming that rebuilds soil health instead of depleting it, through living roots, covered soil, minimal disturbance, crop diversity, and feeding the soil biology rather than only the plant.

Is regenerative agriculture the same as organic?

No. Organic stops using synthetic inputs; regenerative builds soil while doing it. A farm can be certified organic and still lose topsoil every season. Many regenerative farms are organic, but organic certification alone does not require regeneration.

Can a small farm in the Philippines do this?

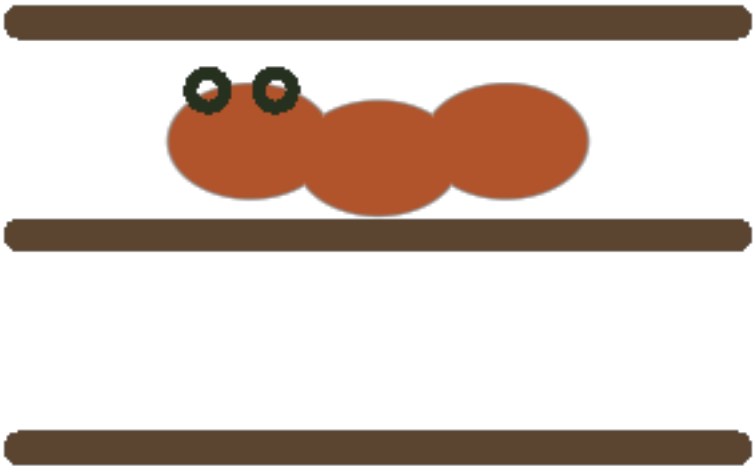
Yes, and it is often easier than on a large one. The method is built around inputs the farm already produces, rice straw, manure, crop residues and on-farm seed, which lowers cost rather than raising it, and it is timed to the wet and dry seasons rather than a temperate calendar.

Does regenerative agriculture cost more to start?

Usually less. Most of the five practices replace purchased inputs with things the farm already grows or throws away. The real cost is attention and timing, which is what the course teaches, and the first buyers come in at a Founding Farmer rate in exchange for honest feedback from their own land.

Chapter 2. Know Your Soil

How to rebuild living soil: what to feed the biology, what to stop doing, and the four tests that prove it is working.



Short answer. Soil regeneration is rebuilding the living part of your soil: the biology, the structure, and the organic matter that hold water and feed crops. You do it by keeping roots growing, keeping the ground covered, disturbing it less, and returning farm residues to the land instead of burning them.

The soil is not a container. It is a living system you can feed.

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What actually lives in your soil

A single handful of healthy farm soil contains more living organisms than there are people on the planet: bacteria, fungi, protozoa, nematodes, earthworms, and the roots themselves. Those organisms are not decoration. They are the mechanism that turns straw, manure, and dead roots into the nutrients a crop can use.

When farmers talk about feeding a plant, the honest version is feeding the biology and letting the biology feed the plant. A bag of urea delivers nitrogen directly to the root. Compost and residues deliver the same thing slowly, by feeding the organisms that make it available, and those organisms also build the glue that holds soil particles into crumbs.

That structure is the part conventional farming quietly loses. Repeated tillage and bare fallow break the crumbs apart, the biology starves without roots to feed on, and the soil compacts into something that sheds water instead of absorbing it. Regeneration is the process of reversing that, in order.

The four things that rebuild soil, in priority order

Keep a living root in the ground as much of the year as possible. Roots are the biology's food supply, delivered underground where it is needed. Bare soil has no food supply. This is the single highest-impact change and it costs a packet of cover crop seed.

Keep the surface covered. Straw, cut residue, or a living cover. Cover holds moisture through the dry season, stops the wet season's rain from hammering bare ground, and suppresses weeds while it works.

Stop the disturbance. Every pass of a rototiller slices fungal networks and releases carbon to the sky. Reducing tillage is not about laziness; it is about not undoing the other three items every season.

Return what the farm produces. Rice straw, animal manure, weeds before they seed, kitchen waste, crop residues. On this farm, rice straw used to be burned in the field. Now it becomes compost and mulch, and it is the cheapest fertilizer on the property.

Notice the order. Farmers who start with the fancy inputs, teas, and amendments usually fail, because the biology those inputs are meant to feed is not there yet. Bare soil needs cover before it needs a brew.

Simple tests a farmer can run, with no lab

You do not need a laboratory to know whether soil is improving. Four tests, done with hands and household items:

The worm count. Dig a spade of soil and count the earthworms. Zero is a warning. Two or three per spade is workable. Eight or more is a working soil. Do it in the same spot each season, and the trend is your measure.

The jar test. A handful of soil in a jar of water, shaken, then left overnight. Sand settles first, silt next, clay last. The proportions tell you how your soil holds water and how to time irrigation.

The infiltration test. Push a tin can open at both ends into the ground, pour in water, and time how long it takes to disappear. Slow infiltration is compaction, and compaction is the problem cover crops and worms fix.

The smell and the crumb. Healthy soil smells like the forest floor. Degraded soil smells flat or sour. Squeeze a clump: healthy soil crumbles into crumbs; degraded soil either shatters into dust or packs into a hard lump.

Write the results in a notebook with the date and the field. Those four lines, repeated year over year, are a better record of your farm than any soil test you would have to pay for.

What changes month to month in the tropics

Temperate soil advice assumes a frozen winter pause. There is no pause here. The wet season accelerates everything, including the loss of your topsoil if it is bare, and the dry season bakes whatever you leave exposed. The practical rhythm that works on our farm:

Wet season: keep the soil covered above all. This is when rain does the most damage to bare ground and when compost breaks down fastest. Apply compost before the rains, not during.

Dry season: mulch is your irrigation. A straw layer on the beds can cut the number of waterings noticeably in the hottest weeks, and it keeps the biology alive under the surface when the topsoil dries.

Between crops: never leave the ground empty. A fast cover crop or a mulch layer is the difference between soil that keeps improving and soil that resets to zero.

The three things to stop doing

Most soil damage on a working farm comes from three habits, and each is a saving when it stops:

Burning residues. Rice straw and crop trash burned in the field are free fertilizer going up in smoke, and the burn also kills the surface biology. On our farm the straw now goes to compost and mulch. Nothing was bought to make that change.

Leaving the ground bare between crops. The fallow period is when rain and sun do their worst to unprotected soil. A fast cover crop or a mulch blanket over the gap costs little and holds the gains you already made.

Tilling out of habit. Tilling once to break ground for a new bed is a decision. Tilling every season because it is what the farm has always done is a recurring bill paid in soil structure, fuel, and the biology underneath.

None of the three costs money to change. Each one stops a loss that is already happening on the farm, which is why the transition usually shows up in the budget before it shows up in the soil test.

Organic matter, in plain numbers

Organic matter is the part of the soil that used to be alive: decomposed roots, residues, manure, and the organisms themselves. It is what gives soil its dark colour, its crumb structure, and most of its water-holding ability. A soil with more organic matter needs less irrigation and less bought fertilizer for the same crop.

The practical implication is that you cannot buy your way to it quickly. Compost and mulch add some each season, roots add more, and the total moves slowly in the tropics because heat breaks material down fast. What a farmer can do is stop the losses, add material every season, and measure the direction rather than the number.

When you will see results

Honest timelines, because they matter for decisions: the worm count and infiltration improve first, often within one or two seasons, because the cover crop is feeding the biology immediately. The smell and the crumb, within a year. Visible darkening of the topsoil and better water holding take longer, usually two to three years in tropical conditions, because organic matter builds slowly even though it breaks down quickly here.

What shows up first in the wallet is different again: input costs fall in the first year, as compost and residues replace a portion of purchased fertilizer. That is the part that keeps farmers in the transition long enough for the soil to answer.

Where this module sits in the method

Soil is module two of ten, right after the case for changing and right before compost. The order matters: you cannot judge whether your compost is working until you can read the soil that receives it. Related pages: compost and biofertilizers , cover crops , stopping tillage , and what regenerative agriculture means .

Frequently asked questions

How long does it take to regenerate soil?

The first signs, worms and water infiltration, often show within one or two seasons. The smell and crumb structure improve within a year. Visible topsoil darkening and real water-holding capacity take two to three years in the tropics, because organic matter builds slowly even though it breaks down fast in the heat.

Do I have to stop using all fertilizer?

No. The method reduces purchased inputs as the farm's own compost and residues take over that job, which is different from banning them. Most farmers cut a portion in the first year and adjust from there, because the reduction is the economic reason to do this at all.

What if my soil is already hard and compacted?

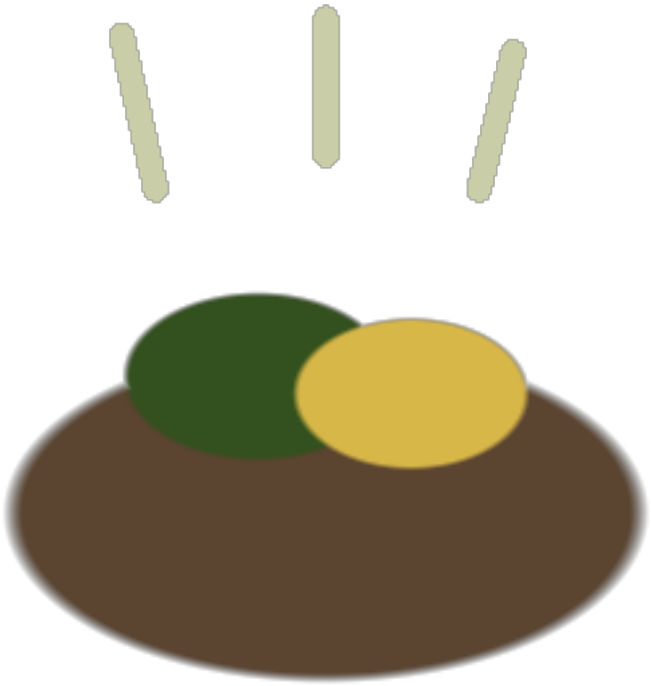
Compacted soil is the most common starting point and it responds well. Start by growing something with a deep root, like a daikon-type radish or a legume cover, to break the layer biologically, keep the surface mulched, and let worms do the rest. Avoid tilling it again to fix it, which usually makes compaction worse.

Can I do this on a very small plot or a garden?

Yes, and it happens faster at small scale. A backyard bed with added compost, a mulch layer, and no digging will show worm activity and better structure within a single season, which makes a garden a good place to learn the method before applying it on a field.

Chapter 3. Compost and Biofertilizers

Fertility from what the farm already produces: rice straw, manure, kakawate, coir. The module that pays for the course fastest.



Short answer. Regenerative farming replaces most bought fertilizer with fertility the farm already produces: compost from rice straw and animal manure, fermented biofertilizer from plant material, and mulch. The biology feeds the crop, input costs fall in the first year, and every ingredient is something the farm was already growing or throwing away.

The farm already grows its own fertilizer. Most of it gets burned.

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Why the farm's own materials come first

A rice farm throws away its own fertility every season. The straw is burned or hauled off, the manure from animals is left to wash away, and the weeds are composted by the sun on a pile at the field's edge. The same farm then buys fertilizer to replace what it threw away. That loop is the single biggest input cost on most small farms, and it is the easiest one to close.

The materials a Philippine farm already produces cover most of what its crops need:

Rice straw , the biggest volume of all: carbon for the compost pile and the best mulch on the farm. Burned, it is fertilizer going up in smoke.

Animal manure from chickens, goats, and pigs: the nitrogen that composts the straw. A farm with animals has the complete compost recipe already on the property.

Kakawate and madre de cacao leaves , the classic Philippine green manure: cut the branches, wilt the leaves, chop them in. Every old boundary fence here is already planted with them.

Coconut coir and husks , slow to break down and excellent for structure, holding moisture in sandy beds for years.

Kitchen waste, weeds before they seed, and crop residues : the small streams that keep the pile running between big additions.

Notice what is not on the list: anything bought. That is the point. The method's compost module is not about buying a premium input; it is about the fertility loop the farm already has, closed.

Compost, the core recipe as a principle

Composting is a managed rot: the right balance of carbon and nitrogen, moisture, and air, and the biology does the rest. The principle that works on a tropical farm:

Layer roughly two to three parts carbon (straw, dry leaves, coir) to one part nitrogen (manure, fresh green material). Too much straw and the pile sits cold for months; too much manure and it turns to sludge and smells. The layering is the balance.

Keep it moist as a wrung-out sponge. In the dry season this means watering the pile; in the wet season, covering it. A pile that dries out stops; a pile that waterlogs goes anaerobic and sour.

Give it air. Turn the pile when the center cools, roughly every week or two in the tropics, where heat and biology work fast. Or build the pile loosely enough that air moves without turning.

Let it cook, then let it rest. In tropical heat a well-built pile breaks down in weeks rather than months. It is done when it smells like forest floor, is dark and crumbly, and you cannot recognize the straw.

The tropical advantage is real: warmth and moisture mean compost happens fast here, once the pile is built right. The tropical risk is equally real: a pile left uncovered in the wet season leaches its nutrients into the ground. Cover the pile in the rains, water it in the dry.

Biofertilizer: the fermented liquid, honestly explained

Biofertilizer, in the form Filipino farmers brew it, is a fermented extract of plant material and animal manure in water. The principle: pack fresh green material (kakawate leaves, weeds, azolla, banana stems) with a smaller amount of manure into a container, cover with water, add a sugar source (molasses or muscovado) to feed the fermentation, and let it ferment for weeks with the container loosely covered, releasing the gases daily.

What it does, stated honestly: the finished liquid carries the soluble nutrients from the material it was brewed from, plus the biology of the fermentation itself. Applied to soil or as a spray on the ground around crops, it feeds the soil biology the same way compost does, in liquid form. It is not magic and not a pesticide. It is the farm's fertility, made mobile.

Two honest cautions. First, brewing recipes vary widely by region and by who taught you, and the ratios above are principles, not a certified formula; a farmer learning this from their local agriculture office or their neighbors should use their version. Second, apply fermented liquids to the soil around the crop rather than onto leaves in the heat of the day, and always from a well-fermented, non-foul batch. A sour, putrid brew went anaerobic; compost it instead.

How and when to apply

Compost goes on before planting and as a top dressing. Worked lightly into beds at planting, or laid on the surface around growing crops and let the worms take it down. Compost is not dug in deeply; feeding from the top is how the biology expects it.

Mulch and compost stack. Compost on the soil, straw mulch on top. The mulch keeps the compost's moisture in and feeds the surface as it breaks down.

Biofertilizer feeds during the season. Applied around the root zone during the crop's growth, when the compost's slow release is not enough on its own.

Timing follows the seasons. Apply compost before the wet season, so the rains work it in; go easy in the peak of the dry unless the beds are mulched and watered.

What it replaces, in pesos

This is the module that pays for the whole course fastest, and the arithmetic is the reason. Every sack of rice straw converted to compost or mulch is a sack of fertilizer not bought. Every pile of manure composted is nitrogen the farm does not purchase. On our farm the loop closed like this: straw to mulch and compost, manure into the pile, kakawate from the fence line chopped in green. Fertilizer purchases dropped from every season to occasional, for specific needs, and nothing was bought to make the change.

Honest limits: compost is not a complete substitute for every nutrient in every crop, and a soil that is genuinely deficient in something specific still needs that something. The measurement module covers how to tell the difference between a biology problem and a mineral one. The transition usually replaces most of the fertilizer bill in the first year, not all of it, and that is still the fastest return of any module in the method.

Starting the first pile

The first pile is the whole transition in miniature, and it is deliberately small. A pile of about a cubic meter, built in one afternoon from materials on the farm: straw layered with manure, watered as you build, left to cook. The point of starting small is that mistakes are cheap. A first pile that goes anaerobic, sits cold, or dries out teaches the balance better than a lecture, and none of those failures cost anything but a turn of the fork.

Walk the pile every few days: feel the center's heat, smell it, watch for the settlement that says it is working. By the second or third pile the balance is in your hands rather than in a recipe, and that is the actual skill: a farmer who can read a pile can make fertility anywhere on the farm, from whatever the season left behind.

How it connects to the rest of the method

Compost is module three of ten, right after knowing your soil and before stopping tillage , because the pile should be running before the last tiller pass. It feeds directly into the cover crops and the economics , and the full system is taught in The Method course .

Frequently asked questions

How long does compost take in the tropics?

Weeks rather than months, once the pile is built right. Tropical heat and moisture make the biology work fast: a well-built pile with the right carbon to nitrogen balance, kept moist and turned, is dark, crumbly, and unrecognizable from its ingredients in a few weeks.

Can I compost rice straw directly?

Yes, with manure or fresh green material to balance it. Straw alone is almost pure carbon and sits cold for months. Layered two or three parts straw to one part manure, kept moist and turned, it breaks down fast, and the pile is the cheapest fertilizer a rice farm can make.

What is biofertilizer made from?

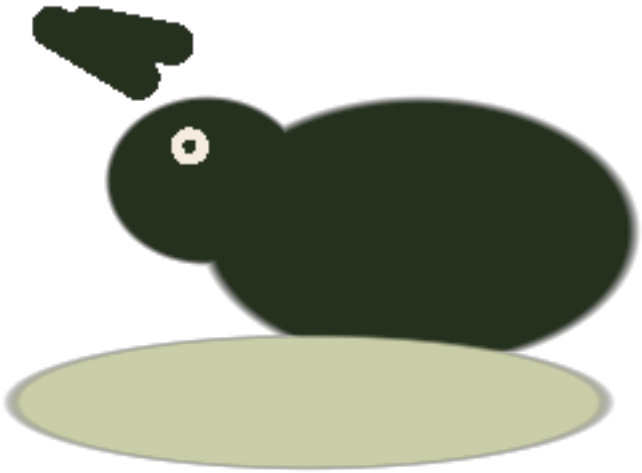
Fresh green plant material, a smaller amount of animal manure, water, and a sugar source like molasses, fermented together for weeks. On Philippine farms the plant material is usually kakawate or madre de cacao leaves, weeds, azolla, or banana stems, all of which the farm or its boundaries already produce.

Can compost replace all bought fertilizer?

Most, usually, in the first year or two. Compost and biofertilizer replace the bulk of the recurring fertilizer bill, but a genuinely deficient soil still needs its specific missing nutrient. The measurement module covers how to tell a biology problem from a mineral one, so you only buy what the farm actually lacks.

Chapter 4. Stop Tilling

Minimal disturbance adapted to rice paddies and vegetable beds, and what happens to structure and weeds when you stop.



Short answer. No-till farming means disturbing the soil as little as possible: no routine rototilling, no plowing out of habit. Every pass breaks fungal networks and releases carbon, so the gains from compost and cover crops are undone each season. You stop by keeping the soil covered, opening beds by hand, and letting biology do the work tillage used to do.

Every pass of the tiller undoes a season of good work.

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What tillage actually does underground

Tillage looks harmless from above: a neat, fluffy seedbed, ready for planting. Underground it is a demolition. The rototiller's tines slice through the fungal networks that thread the soil, the same networks that carry water and nutrients to roots. It collapses the crumb structure that lets air and water move. It buries the surface biology in the dark and exposes the buried biology to the sun. And it releases a pulse of carbon to the air, because the sudden rush of oxygen makes the bacteria burn through organic matter that would have lasted years.

The fluffy seedbed itself is the tell. That fluffiness is the structure being broken, and it settles into compaction within a season, which is why tilled soil needs tilling again. The farm ends up on a treadmill: till, compact, till harder, compact more. Every pass feels like preparation and costs more

than it prepares.

What happens when you stop

The first season after stopping is the uncomfortable one, and it is honest to say so. Without tillage keeping the surface loose, crusted or compacted ground stays that way until biology and roots reopen it. Weeds that tillage used to bury arrive on the surface and have to be handled by mulch and competition instead. The transition usually looks like this:

First weeks: nothing looks better. The soil is as it was. This is the observe period, not the failure period.

First season: the surface may crust where it was already degraded, and weed pressure feels higher because you are seeing weeds tillage used to hide. A thick mulch layer handles both.

Second season: worms and roots have started reopening the topsoil. The infiltration test improves. Weed pressure drops as the mulch and cover crops outcompete germination.

Year three: the soil holds its structure on its own. Beds stay open between seasons. The need for any disturbance keeps falling, which is the point.

How to transition without losing the crop

Going cold turkey on degraded soil is the mistake that gives no-till a bad name. The transition that works on a tropical farm:

Cover first. Before you stop tilling anything, get mulch or a cover crop onto the surface. The cover is what replaces tillage's weed control and moisture work. Stopping tillage on bare soil is how you get both a crust and a weed flush.

Stop one field, not the whole farm. Pick the field or bed with the best soil, or the smallest one, and make it the test. The rest of the farm keeps running while you learn what the transition looks like on your own ground.

Open beds by hand. A hoe or a spade, opening a planting hole or a strip, disturbs a fraction of what a tiller does. Hand work is not a romantic choice on a small farm; it is a precision one.

Use the paddy's own cycle. Rice is the exception that helps: a paddy is flooded for half the year, which suppresses weeds biologically. Lowland rice farmers can reduce tillage between crops and let the flood plus a straw mulch do the work, which is a smaller change than in a dryland system.

Let the residues lie. Previous crop's straw and trash stay on the surface as mulch. Direct seeding or transplanting through that residue is the no-till mechanic, and it gets easier every season as the residue layer builds.

The honest exceptions: when tilling is still the right call

No-till is a direction, not a religion, and three cases justify a pass:

A hardpan or plow layer. Compacted layer deep enough that roots and water cannot get through sometimes needs mechanical breaking, once. A deep-rooted cover crop like radish or pigeon pea can do the same job biologically, but it takes seasons. Breaking the pan once, then covering the ground and never doing it again, is the trade that works.

A brand-new bed in degraded ground. Opening a first bed in soil that has been dead for years is a legitimate single pass, paired immediately with compost and a cover so the bed starts regenerating the day it is opened.

A rescue. A soil that has gone anaerobic, a salt problem, a spot that needs regrading. Rare, real, and better handled with one honest pass than with a dogmatic rule.

The difference between these and routine tillage is that they are decisions, made once, with a regenerative plan attached. The treadmill is not a decision; it is a habit.

The first season without a tiller, on our farm

The transition looked like this on our own ground, and it was not smooth. The first season after the last routine pass, the beds crusted where the soil had been tilled for years, and the weeds that used to be buried arrived in force. The mulch layer, rice straw over every bed, was what held the line: it kept the crust from baking and shaded out most of the germination while the biology underneath restarted.

By the second season the difference was measurable rather than visible: the worm count in the beds went up, water stopped pooling the way it used to, and the hoe came out half as often. What had looked like doing nothing was the biology and the cover doing the work the tiller used to do, without the fuel or the damage. The third season, the beds held their structure on their own between crops, and preparation became a hand job rather than a machine job.

The honest lesson from that transition: the first season tests your patience, not the method. A farmer who judges no-till by its first months will quit; a farmer who judges it by its second year keeps the farm.

What replaces each tillage job

Most farmers keep tilling because it does four jobs, and each job has a replacement:

Weed control: replaced by mulch, cover crops, and competition. A thick straw layer suppresses germination better than a hoe, and it works every day instead of once a pass.

Moisture work: replaced by the mulch layer as well. Tilled bare soil loses moisture fast; covered soil keeps it.

Seedbed preparation: replaced by hand-opened holes and strips, and by soil that holds its structure between seasons so there is less to prepare.

Incorporating fertilizer: replaced by surface compost and mulch, which feed from the top down the way the biology expects. Tilling compost in buries the biology that needs it near the surface.

What the neighbors will say

Honest, because it stops more transitions than any agronomic problem: a fallow-looking field with mulch on it draws comment in a barangay where every farm is tilled bare. The neighbors will ask if the farm has been abandoned. Some will tell you the straw should be burned. The answer that works is the field itself, once the second season shows what the soil is doing, and until then the only thing that matters is your own notebook: the worm counts, the infiltration times, and the input costs that move while the farm looks untidy.

This is also why the barangay Farmers Association work matters. A farmer transitioning alone argues with everyone; a group of farmers comparing notes argues with results. The association is where those notes get compared, and it is the reason the method is taught publicly at all.

How it connects to the rest of the method

No-till is module four of ten, right after compost and biofertilizers and before cover crops . The order matters: a cover crop and a compost pile should already exist before the last tiller pass, because they are what replace it. Start at the definition if you are new, and see the first-season plan for the transition laid out as a year.

Frequently asked questions

Do I have to stop tilling completely?

No. The goal is minimal disturbance, not zero. A one-time pass to break a hardpan or open a first bed in dead ground is a legitimate decision. What the method removes is routine tillage, the seasonal pass that exists because it always has.

Will weeds get worse when I stop tilling?

They get more visible in the first season, because tillage used to bury them, then they get better as mulch and cover crops suppress germination. The transition's weed flush is temporary; a farm with a thick permanent cover has less weed pressure than a tilled one within two seasons.

Can a rice paddy be farmed without tillage?

Partially, and the flood helps: a paddy is submerged for half the year, which suppresses weeds biologically. Between crops, keep the straw on the field as mulch and reduce the passes between cycles. Lowland rice is one of the easier systems to transition because the flood does part of the work.

What do I do about hard, compacted soil without tilling it?

Two routes: a deep-rooted cover crop like radish or pigeon pea to break the layer biologically over a couple of seasons, or one honest mechanical pass to break the pan, immediately followed by permanent cover. Both work; what matters is that it happens once, not every season.

Chapter 5. Cover the Soil

Cover crops and the seven-layer planting: canopy, vines, cash crops, covers, roots, groundcover, animals.



Short answer. A cover crop is something you grow to feed the soil rather than to sell. On a tropical farm it keeps living roots in the ground between cash crops, holds moisture, smothers weeds, and adds nitrogen for free when you cut it. Legumes like mung bean and cowpea are the cheapest start, and rice straw mulch does the same job without planting.

Something growing, all year. Even where nothing pays.

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What a cover crop actually does

A cover crop is a plant grown for the soil's benefit rather than for harvest. Between cash crops, on paddies' edges, under trees, and on any ground that would otherwise sit bare, it does four jobs at once:

It keeps a living root in the ground. Roots are the food supply for the soil biology, delivered underground where that biology lives. Bare soil starves it; a growing cover feeds it every day it is in the ground.

It covers the surface. Leaves intercept the rain before it hammers bare ground and shades the soil before the sun bakes it. Covered soil loses less moisture in the dry season and less topsoil in the

wet one.

It competes with weeds. A thick cover occupies the space, light, and nutrients a weed seedling needs. Suppressing germination is cheaper than fighting an established weed.

Legumes add nitrogen. Mung bean, cowpea, and their relatives host bacteria in their roots that pull nitrogen from the air. Cut the crop at flowering and that nitrogen is in the soil, for the price of a packet of seed.

Those four jobs are why cover cropping is the fastest of the regenerative practices to prove itself. The soil responds within a single season, because the roots and the cover are working immediately.

The species that work in the Philippines

The imported guides list species that either do not grow here or behave differently in a wet and dry season cycle. These are the ones that work on tropical ground, and on our farm:

Legumes, the nitrogen builders

Mung bean (munggo). Fast, cheap, familiar to every Filipino farmer, and it fixes nitrogen well. Sow it between rice crops or on fallow paddies, cut it at first flowering, and leave the residue where it falls.

Cowpea. Tougher than munggo in dry conditions, deep-rooted, and it covers ground fast. The seed is edible, so a farmer can let a strip go to harvest and still use the rest as a cover.

Pigeon pea (kadyos). A perennial legume that doubles as food. It grows tall enough to shade out weeds on edges and boundaries, and it keeps producing for years with no replanting.

Crotalaria. Less familiar, but the fastest biomass builder of the set, and it suppresses certain root nematodes. Cut it before it seeds or it will volunteer everywhere.

Grasses and residues, the soil holders

Rice straw mulch. Not planted at all: the previous crop's residue laid over the beds. It is the cheapest cover on any rice farm because it already exists, it suppresses weeds, holds moisture, and breaks down into the same organic matter a cover crop would have made. On our farm, straw used to be burned. Now it is the default mulch on every bed.

Napier and other grasses. On boundaries and slopes, grass strips hold soil against the wet season's rain and give cut fodder for animals. Their root systems are deep and fibrous, which makes them the best erosion stoppers of the set.

Kakawate and madre de cacao. Not ground covers but tree legumes whose leaves make the classic Philippine green manure. Cut the branches, wilt the leaves, and lay them over beds or chop them into compost. Every old farm boundary here already has them planted.

When to cut, and why timing decides everything

A cover crop cut too late becomes a weed. Cut too early and it has not built the nitrogen or the biomass. The rule that works:

Cut legumes at first flowering. That is the moment the nitrogen in their roots peaks. Flowering also marks the point where the plant starts putting energy into seed rather than into roots, so every day after first flowering the soil gains less.

Cut grasses and non-legumes before they seed. Once a grass puts up a seed head, the cover becomes next season's weed problem. Cut it while it is still vegetative and leave the residue on the surface.

Leave everything where it falls. The residue is the point. Cutting and removing a cover crop gives the farm nothing but the work; cutting and leaving it gives the soil the nitrogen, the organic matter, and the mulch layer in one motion.

Never let a cover crop compete with a cash crop. On beds, plant covers between seasons, not beside a growing cash crop. Under trees, keep a ring clear around each trunk and let the cover run to the drip line.

The seven layers: where covers sit in the whole system

A cover crop is one layer of seven on a regenerative tropical farm. The full planting, from top to bottom: canopy trees, vines, cash crops, cover crops, roots, groundcover, and animals in the same system. The layers stack because each one uses a different slice of the light, the space, and the soil depth, which is how a small farm ends up producing more than its area says it should.

On our farm the vine layer is dragonfruit, the cash crop that pays while the other layers build soil. The cover layer feeds everything under it. A farmer does not need all seven layers in the first season; the cover layer is the one to establish first, because it is the cheapest and it makes every layer above it cheaper to run.

What a cover costs, and what it replaces

The economics are the reason this practice comes early in the method. A packet of munggo seed costs less than one bottle of herbicide, and it replaces the herbicide's job plus three others. The comparison that decides it:

Against herbicide : a cover suppresses germination every day it grows, for one purchase, instead of a spray per flush.

Against bought fertilizer : a legume cut at flowering adds nitrogen and biomass for the price of seed, against a bag of urea per season.

Against irrigation : a mulch layer or cover holds moisture through the dry weeks, which is water and fuel the farm does not spend.

Against labor : weeding is one of the biggest labor costs on a small farm. A thick cover removes most of the germination before it becomes a hoe's worth of work.

None of those replacements is a compromise. Each one is the same job done by a plant instead of a purchase, which is why the transition usually pays for itself in the first season it is tried honestly.

Common mistakes, and the fix

Letting a cover go to seed. One season of uncut crotalaria becomes three seasons of volunteer crotalaria. The fix is a calendar: mark the cutting date when you sow, and walk the strip weekly as flowering starts.

Cutting and removing the residue. A removed cover crop is work for nothing. Always leave the biomass on the surface where it grew.

Planting a single species everywhere. One species feeds one pest and uses one soil slice. Mixing a legume with a grass covers both jobs and hedges the failure of either.

Expecting a cover crop to fix compacted ground alone. Compaction needs a deep root plus time; a shallow-rooted cover helps but will not break a hardpan. Pair it with the minimal disturbance approach and give it two seasons.

How it connects to the rest of the method

Cover cropping is module five of ten, sitting between compost and biofertilizers and pests without poisons . It depends on reading your soil to choose the right species, and it feeds the weed strategy in the pests module, because a thick cover is the cheapest weed control a farm has. The full system is in The Method course .

Frequently asked questions

What is the best cover crop for a tropical farm?

Mung bean is the best first cover crop for most Philippine farms: it is cheap, fast, fixes nitrogen, and every farmer already knows how to grow it. Cowpea for drier ground, pigeon pea for boundaries, and rice straw mulch wherever a crop's residue already exists.

When should I cut a cover crop?

Cut legumes at first flowering, when the nitrogen in their roots peaks, and cut grasses before they put up a seed head. Leave all the residue where it falls. A cover crop left uncut past flowering becomes next season's weed.

Do cover crops steal water from my cash crop?

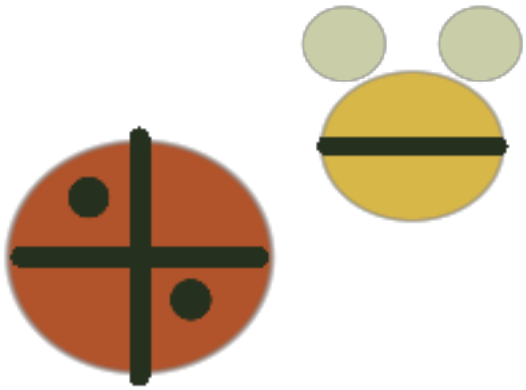
Not when they are managed as covers rather than competitors. Plant them between cash crop seasons or to the drip line of trees, keep a clear ring around each crop, and cut them before the cash crop's water demand peaks. A mulch layer holds more moisture than it uses.

Can I use rice straw as mulch instead of planting a cover?

Yes, and on a rice farm it is the cheapest start. Straw mulch covers the soil, suppresses weeds, holds moisture, and breaks down into organic matter. It does not add living roots while it sits, so pair it with a legume cover on the next gap for the full effect.

Chapter 6. Pests Without Poisons

Why a diverse, covered system gets fewer pest problems, and what to do when they still arrive.



Short answer. Regenerative farming does not use routine pesticide spraying. It prevents most pest problems instead: diverse crops, covered soil, and healthy biology grow stronger plants and feed the insects that eat the pests. When an outbreak still arrives, the response is scouting, then targeted homemade sprays, and only rarely a targeted intervention, used as a decision rather than a habit.

A pest problem is usually a system problem wearing a bug's face.

Short answer Regenerative farming does not use routine pesticide spraying. It prevents most pest problems instead: diverse crops, covered soil, and healthy biology grow stronger plants and feed the insects that eat the pests. When an outbreak still arrives, the response is scouting, then targeted homemade sprays, and only rarely a targeted intervention, used as a decision rather than a habit.

So does regenerative farming use pesticides?

The honest answer: not routinely, and not as the system's tool. A regenerative farm's pest management is prevention first, and the prevention is structural. When an outbreak still breaks through, the response escalates in steps: handpicking and physical removal, then sprays made from farm materials, and in a genuine emergency a targeted intervention on one affected area. What it never reverts to is the calendar spray, the prophylactic routine that exists because it always has.

That distinction matters because the question is usually asked by a farmer deciding whether to change, and the honest framing is this: a regenerative farm is not a farm that never loses anything

to pests. It is a farm where pest losses are the exception rather than the plan, where the response starts with something cheaper than a bottle, and where the system itself does most of the defending.

Why a regenerative system gets fewer pest problems

Four mechanisms, all of them consequences of practices described in other modules:

Diversity starves a specialist. A monoculture is an unlimited buffet for the one insect that eats that crop. A farm growing many crops, in rows and in layers, means no pest finds an unbroken meal. This is the single biggest mechanism and it comes from diverse planting .

Healthy plants defend themselves. A plant fed by living biology builds tougher cell walls, thicker leaf surfaces, and its own chemical defenses. A plant force-fed soluble nitrogen grows soft, fast tissue that sucking insects prefer. This comes from compost and biofertilizer .

The pests have enemies. Ladybugs, lacewings, spiders, parasitic wasps, birds, and frogs eat pest insects, and they live where there is habitat: hedgerows, cover, mulch, and flowers. A farm with habitat has pest control living on it for free. A bare monoculture has nowhere for them to stay.

Healthy soil grows resilient roots. Stressed plants from compacted, waterlogged, or dead soil are the first to attract pests. Much of what looks like a pest problem starts as a root problem. This comes from the soil work .

What to do when pests still arrive

They will still arrive. The escalation that works, in order:

Scout first, every time. Walk the affected area and identify what is actually eating what. Most damage is from a handful of culprits, and each has a different response. Spraying before identifying is the habit the method removes.

Handpick and remove. On a small farm this is genuinely practical for the big culprits: beetles, caterpillars, snails, egg clusters under leaves. A morning's picking beats a season's spraying for the pests it catches.

Homemade sprays from farm materials. The classics that work: chili-garlic-soap sprays against soft-bodied insects, neem where the tree is available against chewing and sucking pests, wood ash as a barrier against some crawlers. These are contact tools: they work where they land, they break down fast, and they do not leave residue on the crop.

Water and timing. A strong jet of water removes aphid colonies. Spraying in the evening, when pollinators are not flying, protects the insects you want to keep.

A targeted intervention, rarely, as a decision. A genuine outbreak threatening a season's work is sometimes answered with a specific, targeted treatment on one affected area, chosen because the alternative is losing the crop. What makes that a decision rather than a habit is that it is rare, targeted, and followed by asking why the outbreak happened, which is usually a system answer: too much of one crop, stressed plants, or habitat that disappeared.

Weeds: indicators first, enemies second

Weeds get the same treatment as insects: understand what they are saying before fighting them. Weeds are honest indicators of soil condition, and a farmer who reads them learns their fields faster than any test:

Compacted, wet ground grows the weeds that tolerate it. Patches of them in a field point at a drainage or compaction problem before the crop shows it.

Bare, depleted ground grows fast, short-lived pioneer weeds whose job is covering exposed soil. They are the soil's emergency response, and they are telling you the farm needs a planted cover instead.

Fertile, healthy ground still grows weeds, but different ones, and slower. The weed pressure drops as the cover crops and mulch take over the space the weeds were filling.

Practically: weeds are managed by competition and cover, the same as insect pests are managed by habitat and diversity. A thick mulch layer, the cover crops , and never leaving ground bare do more for weeds than any spray, because they remove the space the weeds were colonizing.

Building the habitat that holds the good insects

The pest predators are only resident if the farm has somewhere for them to live. Habitat is the part of pest control that costs nothing and gets skipped most often:

Hedgerows and boundary plantings. The kakawate and madre de cacao on the fence lines are not just green manure: they are perches for birds, cover for spiders and frogs, and nectar for the parasitic wasps that lay their eggs in caterpillars. Every boundary planting is pest control standing still.

Flowers through the seasons. The adults of many pest-eating insects drink nectar; their larvae eat the pests. A farm with something flowering year round keeps the adults resident, which means their larvae are on station when the pests arrive.

Mulch and groundcover. Ground beetles, spiders, and frogs live under the straw layer. A bare farm has no ambush predators; a mulched one has them everywhere.

A water source. Even a small pond or a wet ditch keeps frogs resident, and frogs are among the best pest control a tropical farm has.

On our farm, the chickens and goats are part of this too, managed so they clean up fallen fruit and scratch the ground behind the animals' rotation without touching the crops. The animals are a layer of the system, and pest cleanup is one of their jobs.

Disease: the same answer with a longer fuse

Fungal and bacterial disease follows the same logic: prevention is structural, and the structure is air, light, and dry leaves. Crowded plantings hold humidity on leaf surfaces; the tropics grow fungus fast in still, wet air. Spaced plantings, pruned canopies, and morning irrigation (so leaves dry in the day rather than staying wet overnight) prevent most of it. Diseased material is removed and composted hot rather than left to spread, and rotation keeps soil-borne disease from building up in one bed.

How it connects to the rest of the method

Pests are module six of ten, right after covering the soil and before water and seasons . The pest module is where the earlier modules pay off visibly: the diversity, the healthy plants, and the habitat they built are the actual pest control. See what regenerative agriculture means for the definition behind all of it, and The Method course for the full system.

Frequently asked questions

Does regenerative farming use pesticides at all?

Not routinely. Prevention is the tool: diversity, healthy plants, and habitat for pest predators. When an outbreak still arrives, the response starts with handpicking and homemade sprays from farm materials, and a targeted intervention is a rare, deliberate decision on one affected area rather than a seasonal routine.

What is the best homemade pest spray?

Chili, garlic, and soap blended and strained in water is the workhorse against soft-bodied insects, and neem where the tree grows covers chewing and sucking pests. They are contact sprays: they work where they land and break down fast, so they are applied to the pest, in the evening, rather than across the whole farm.

Will I lose more of my crop to pests without spraying?

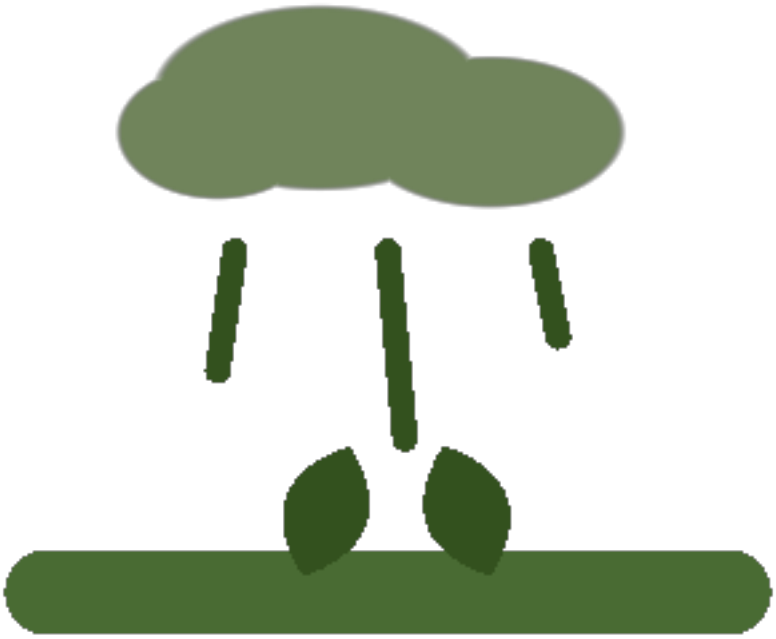
In the transition seasons, possibly, while the habitat and diversity establish. Once the system is running, most farms see pest losses become the exception rather than the plan, because the mechanisms that prevent outbreaks are structural: no unbroken buffet for specialists, tougher plants, and predators living on the farm.

What do weeds tell you about your soil?

Weeds are indicators: patches of wet-ground weeds point at drainage or compaction problems, pioneer weeds on bare ground point at depleted soil that needs a planted cover, and the overall weed pressure drops as mulch and cover crops take over the space. Read them before fighting them.

Chapter 7. Water and Seasons

The tropical wet and dry calendar, typhoon resilience, irrigation timing, and the honest carbon story.



Short answer. Regenerative farming holds more water in the soil because covered, living ground absorbs rain instead of shedding it, and organic matter holds moisture like a sponge. The same biology builds soil carbon slowly and honestly, and covered soil stays on the land through a typhoon instead of washing away.

The wet season is not the problem. Bare soil in the wet season is.

Short answer Regenerative farming holds more water in the soil because covered, biologically active ground absorbs rain instead of shedding it, and organic matter holds moisture like a sponge. That same biology builds soil carbon slowly and honestly, and covered soil stays on the land through a typhoon instead of washing away. The tropical calendar, wet and dry, is the timing that makes all of it work.

How does regenerative agriculture reduce carbon emissions?

The honest answer has two parts, and the second is the one most marketing skips. First: photosynthesis and roots put carbon into the soil, and regenerative practices keep it there. Every growing root leaks carbon to the biology around it, compost and residues add more, and minimal disturbance stops the release that tillage causes. That is real, and it is measurable in the soil's darkening over years.

Second, the honest part: soil carbon builds slowly. A farm regenerating its soil sequesters carbon year over year, in tons per hectare that vary widely by soil, climate, and practice, and it takes years

to become significant. A farm does not offset anything in its first season, and anyone selling fast carbon numbers is selling something else. The reason to do this is that the same practices that build soil carbon also cut the farm's actual emissions: no residue burning, fewer passes of machinery, less fertilizer manufactured and hauled to replace what the farm threw away.

On our farm the emissions that fell first were the obvious ones: the straw stopped being burned every season, the tiller runs a fraction as often, and the fertilizer the farm buys dropped because the compost pile replaced it. Those are real reductions, visible in the budget and in the air over the barangay, and they came from the same work that builds the soil's carbon slowly underneath.

How covered soil holds water

Water is where regenerative farming pays fastest in the tropics, and the mechanism is simple. Bare soil sheds heavy rain: the drops pound the surface, seal it, and the water runs off, taking topsoil with it. Covered soil drinks it: the canopy breaks the drops, the mulch and cover slow the water at the surface, and the crumb structure and channels left by roots and worms let it soak in.

Rain that soaks in stays. Water absorbed into the soil profile is the farm's reserve for the dry weeks. Water that runs off is a loss of both the water and the soil it carried.

Organic matter holds moisture. Soil with more organic matter holds substantially more water for the same rainfall, which is the difference between a crop that rides out a dry spell and one that needs irrigation two days after the rain stopped.

Mulch cuts evaporation. A straw layer on the surface shades the soil and breaks the wind, which is most of the evaporation in the hot weeks. The moisture stays where the roots are.

Living roots keep the channels open. The pathways roots leave behind are how water moves downward instead of pooling. A farm with roots growing year round has drainage that a tilled farm has to buy with machinery.

The tropical calendar: wet and dry, not spring and fall

Every temperate farming book assumes four seasons and a winter that resets the system. The Philippines has two, and the timing that works:

Before the wet season: everything that can be covered, covered. Compost applied and mulched so the rains work it in rather than washing it away. Cover crops established so the ground is never bare when the heavy rains start.

The wet season: the season of loss if the soil is bare, and the season of free fertility if it is covered. Drainage matters now; the swales, ditches, and raised beds that move excess water are checked before the typhoons, not after.

After the typhoon: the covered farm's residue and mulch stay on the land; the bare farm's topsoil is in the drainage ditch. This is where the resilience claim is real, and it is the argument that convinces neighbors faster than any carbon number.

The dry season: mulch is the irrigation. The beds that were mulched before the rains hold their moisture into the dry weeks; the ones that were not need watering from the first hot week. Cover crops that tolerate drought keep living roots in the ground while everything else waits.

Typhoon resilience, stated plainly

A typhoon does three kinds of damage to a farm: it washes bare soil away, it flattens and floods crops, and it takes weeks of work with it. The regenerative system does not make a farm typhoon-proof, and honesty requires saying that. What it changes is which losses happen:

Soil loss: covered soil stays. This is the biggest difference and the most visible after the storm: the covered farm's mulch is still there, the bare farm's topsoil is not.

Flooding: soil that absorbs water drains faster after the storm. Beds that sit high and drain well come back into production weeks earlier than compacted, waterlogged ones.

Deep-rooted perennials hold. The trees, the dragonfruit vines on their posts, and the deep-rooted covers come through wind better than shallow annuals. The seven-layer system means a typhoon that flattens the annual layer leaves the canopy and the vines standing.

Diversity spreads the risk. A storm that takes one crop does not take the farm's income, because the income was never in one basket.

Irrigation timing, as a principle

The method reduces irrigation rather than banning it, and the timing that works in the tropics: water in the early morning, so leaves dry in the day and disease does not get its humid overnight start. Water deeply and less often, so roots follow the moisture downward rather than sitting shallow. Let the mulch do its job between waterings, and let the infiltration and moisture tests tell you when the soil actually needs water rather than the calendar.

What the dry season looks like on our farm

The dry months are when the water work pays, and it is worth describing concretely. The beds that went into the dry season mulched and composted hold their moisture from one watering to the next; the beds that went in bare need water from the first hot week and again every day after. The dragonfruit, deep-rooted and mulched, rides out the dry weeks without irrigation. The cover crops on the fallow ground either tolerate the drought or die back and become the next mulch, which is the system refusing to waste a failure.

The goats and chickens range the drier edges, and their manure lands where the next season's compost will need it. Nothing about the dry season is idle on a farm running this system: the mulch is working, the deep roots are working, the biology under the surface is working, and the calendar is already marking when the wet season's preparation starts. A farm that planned its water work before the rains spends the dry season watching it pay; a farm that did not spends it buying water.

How it connects to the rest of the method

Water and seasons is module seven of ten, after pests without poisons and before the first-season plan , because the calendar is what turns the practices into a year. It depends on the cover layer and the soil structure the earlier modules built. The full system is in The Method course .

Frequently asked questions

Does regenerative farming really reduce carbon emissions?

Yes, in two honest ways: the practices cut the farm's real emissions now, by ending residue burning, reducing machinery passes, and replacing manufactured fertilizer, and they build soil carbon year over year, slowly, through roots, compost, and minimal disturbance. Anyone promising fast carbon numbers is selling something other than soil.

How much water does mulch actually save?

Enough to change the irrigation schedule. A straw layer shades the soil and breaks the wind that drives evaporation, so beds stay moist between waterings instead of baking in the hot weeks. The exact number varies by soil and weather, which is why the farm's own moisture tests, not a chart, are the measure.

Can regenerative farming survive a typhoon?

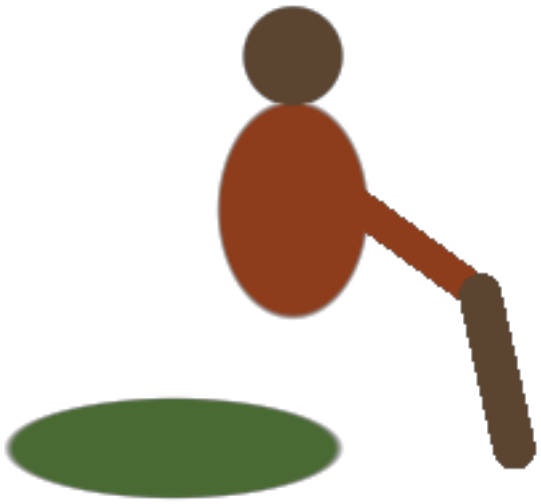
No farm is typhoon-proof, and honesty requires saying that. What the system changes is which losses happen: covered soil stays on the land instead of washing away, well-drained beds come back into production weeks earlier, deep-rooted perennials hold through wind, and diverse crops mean one lost harvest is not the income gone.

When should I plant before the wet season?

Establish covers and apply compost before the heavy rains start, so the ground is never bare when they arrive and the rains work the compost in rather than washing it away. The exact weeks follow your own local calendar, and the module on seasons is about reading your own rain, not a fixed date.

Chapter 8. The First-Season Conversion Plan

A step-by-step first year on your own land: what to do first, what to skip, and the beginner mistakes.



Short answer. Start regenerative farming by observing before changing anything, then in order: cover every patch of bare soil, start one compost pile, stop tilling one field, and write everything down. Skip the big purchases, the whole-farm conversion, and the premium market plans in year one. The first season builds the system; the harvest it improves comes in the seasons after.

How to start without buying your way in or breaking your farm.

Short answer Start regenerative farming by observing before changing anything, then in order: cover every patch of bare soil, start one compost pile, stop tilling one field, and write everything down. Skip the big purchases, the whole-farm conversion, and the premium market plans in year one. The first season builds the system; the harvest it improves comes in the seasons after.

What the first season is actually for

The first season of regenerative farming is not about proving the method works. It is about building the system that makes every season after it work: the cover on the soil, the compost pile running, the notebook started, and one field where you have watched the biology come back. A farmer who spends year one on those four things enters year two with a running system. A farmer who spends year one trying to convert everything at once, buying inputs and chasing a premium market, usually quits by the second, because nothing was ever measured and nothing was ever allowed to establish.

The steps below are in the order they actually happen on a tropical smallholding, and each one is deliberately small. The method's first season is a sequence of cheap, reversible decisions, not a leap.

The first-season steps

Walk the land and change nothing, for the first weeks. Note where water stands after rain, where the soil is bare or crusted, what weeds grow where, what already grows well, and where the wind and the sun hit hardest. Write it down. The urge to act immediately is the first beginner mistake: every change you make before observing is a guess about a farm you have not yet read.

Cover every patch of bare soil. This is the first real action and the cheapest: rice straw, crop residue, or a fast cover crop like munggo, over every exposed patch. Bare soil is losing moisture, biology, and topsoil every day it sits exposed. Covering it is the single highest-impact change on the farm and it costs almost nothing.

Start one compost pile. Small, about a cubic meter, from materials the farm already produces: straw layered with manure, watered as you build, turned when the center cools. One pile, not a composting operation. It teaches the balance cheaply and it is running by the time you need it.

Start a biofertilizer brew. Once the first pile is going, a container of fermented plant material and manure gives the farm its liquid fertility for the season. Kakawate or madre de cacao leaves, weeds, manure, molasses, water, weeks of fermentation. Small batch, from the farm's own boundaries.

Pick one field and stop tilling it. The best-soiled or smallest field, covered first, is the test. Hand-open the planting holes, keep the mulch on, and let the cover crop run between crops. The rest of the farm keeps its old routine while you learn what the transition looks like on your own ground.

Plant the cover on the gaps. Between cash crops, on paddies' edges, under trees: something growing everywhere the cash crop is not. This is the layer that keeps the biology fed and the weeds out, and it is the difference between a covered farm and a regenerating one.

Write down the soil tests and the costs. Worm count, infiltration, smell, crumb: one line each, per field, with the date. Input costs and harvest in the same notebook. This is the step that makes everything else provable, and it is the one skipped most often.

Watch the wet season. If the first season includes the rains: the covered ground holds, the bare ground washes, and the difference is the most convincing argument on the farm. Check the drainage before the typhoons, not after.

Read the results honestly. At the season's end, the notebook tells you what moved: worms, water, weeds, costs. Mostly flat numbers after one season are normal; the direction is what matters, and year two is where it compounds.

What to skip in year one

Just as important as the steps, because these are the mistakes that cost a year:

Big purchases. No equipment, no amendment supply, no brewing hardware beyond a container. Every regenerative practice in the first season runs on things the farm already has. The farm that buys its way in learns nothing and spends what the transition was supposed to save.

The whole-farm conversion. Stopping tillage everywhere, covering everything, and rebuilding all at once produces a weed flush, a workload shock, and a discouraged farmer. One field, done properly, converts the farmer; the farmer converts the farm.

The premium market plan. Selling a premium crop requires a consistent harvest, which requires a running system, which takes seasons. Planning the market before the system produces is how farms end up promising what they cannot fill.

Chasing every practice at once. The seven layers, five crops, compost, brews, and measurement all in year one means none of them established. Two or three done properly beat nine attempted.

Judging by the first harvest. The first season's numbers are usually flat. The farmers who quit do it on year one's feelings; the ones who stay do it on year one's direction, which the notebook holds.

The classic beginner mistakes, and the fix

Tilling the mulch in. The straw layer is doing its job on the surface; burying it restarts the damage. Feed from the top.

Cover crops gone to seed. One uncut strip of crotalaria becomes next season's weed problem. Mark the cutting date when you sow.

Compost that stopped. A pile left to dry out or waterlog in the rains goes cold and stays there. Walk it every few days; water it in the dry, cover it in the wet.

Spraying before scouting. The calendar spray habit survives the transition longer than any tiller. Identify first; the pests module covers the escalation.

Expecting the neighbors to approve. A mulched field looks abandoned to a barangay of bare-tilled farms. The second season's field is the answer, and the notebook is the proof.

The first season on our farm, as it happened

The order above is not theory; it is the order the transition actually ran on our own ground, and the parts that did not go to plan are the useful part. The first mulch went on late, after the first hot weeks had already baked the beds, which is how we learned that covering comes before anything else. The first compost pile sat cold for a month because it was all straw and no manure, which is how the balance became a felt thing rather than a recipe. The first field without tillage crusted and drew comments from every neighbor, which is how we learned that the cover had to be thick enough to answer them.

What went right was smaller than expected and more important: the worms came back in the test field within the first season, the straw stopped being burned, and the notebook, in the father's handwriting, started holding what the farm was actually doing. That notebook is now the record the whole method is taught from, and it started as the least impressive thing on the farm.

The first season did not prove the method. It built the system that proved it in the seasons after, which is exactly what a first season is for.

How it connects to the rest of the method

The first-season plan is module eight of ten, but it is where a farmer starts the whole method in practice. It depends on the soil tests , it runs on the compost and the cover , and it feeds the notebook that never stops. The full course is in The Method , and the Find Your Path quiz tells you which module your farm should start with.

Frequently asked questions

How do I start regenerative farming on a small budget?

With what the farm already produces: cover the bare soil with rice straw or crop residue, start one compost pile from straw and manure, plant a packet of munggo on the gaps, and write everything down. The first season's only purchase is a packet of seed, and the transition was designed so that is enough.

How much land do I need to start?

One field, one bed, or one backyard plot. The method works at any scale and proves itself fastest at small scale, where one person can cover the soil, watch the pile, and read the results without help. Start with the smallest piece of ground you control and let it convert the rest.

How long before a regenerative farm becomes profitable?

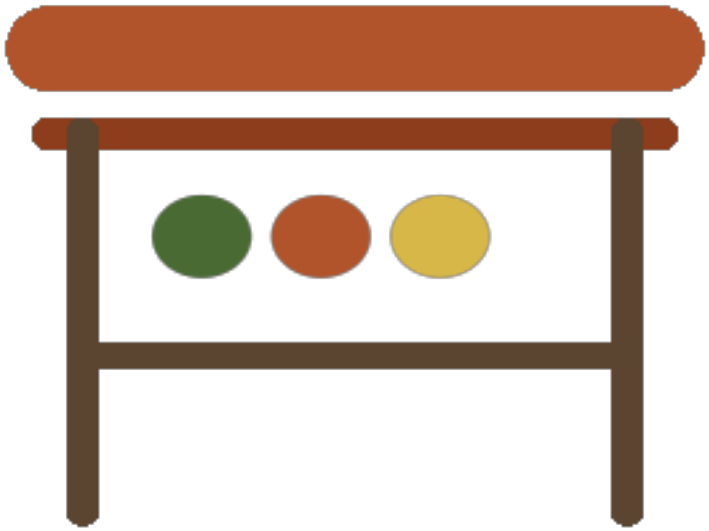
The costs fall first, often in the first year, as compost and residues replace purchases and the tiller idles. The yields stabilize and climb in the second and third seasons as the biology establishes. Anyone promising a first-season profit is guessing; the honest answer is that the budget improves before the harvest does.

What should I NOT do in my first season?

Three things: do not buy equipment or inputs, do not convert the whole farm at once, and do not plan a premium market before the system produces consistently. The first season is for building the system on one field with what the farm already has, and for starting the notebook that proves it worked.

Chapter 9. Making It Pay

Where the money comes from, in order: input costs first, premium sales second, grants covered honestly. Dragonfruit as the cash-crop case.



Short answer. Regenerative farming pays in a specific order: input costs fall first, in the first season, because compost and farm materials replace purchases. Premium sales and direct marketing come second, once the system produces consistently. Grants help but are never guaranteed, and they are country-specific, so check your own agriculture office.

The transition is a business decision. Here is the arithmetic.

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Where the money actually comes from, in order

Farmers are told regenerative farming "pays" without being told when or how, which is why the claim reads as marketing. The honest version is a sequence, and the order matters because each source of income depends on the previous one being in place:

Input costs fall first. Compost from straw and manure replaces a recurring fertilizer purchase. A cover crop replaces herbicide. Mulch replaces irrigation hours. This happens in the first season, it is the largest and fastest source of gain, and it requires nothing bought. The transition pays before the harvest changes, which is what keeps farmers in it.

Labor shifts. Weeding and watering hours drop as mulch and covers take over. The hours that remain move from repeated chores to the season's actual work. On a family farm this is the difference between the farm running the family and the family running the farm.

Risk falls. A diverse farm's income is never in one basket: a crop that fails, a price that crashes, or a typhoon that flattens one layer does not take the year. This is invisible in good years and the whole story in bad ones.

Premium sales come second. Once the system produces consistently, a farm can sell on how it grows: to buyers who pay for verified practice, at markets where the story matters. This is year two or three, not year one, and it depends on the measurement record the farm kept.

Direct and cooperative marketing last. Selling direct, or pooling with neighbors through a cooperative, captures the margin a middleman otherwise takes. This needs volume and consistency, which the system builds, and it is where the transition's economics compound.

Grants, covered honestly

Grants are the topic farmers ask about first and the one to be most careful with, because the honest truth is unglamorous: no grant is guaranteed, most are competitive, and they are country-specific and region-specific, which means the only reliable source is your own agriculture office.

What actually exists, by category, and what each is really for:

Local government agriculture offices. Most Philippine municipalities and cities run farming programs: trainings, seed and input distribution, composting facilities, equipment lending. These are the first stop because they are local, they know your barangay, and the application is a conversation rather than a paperwork mountain. The Farmers Association in your area usually knows what is currently funded.

National agriculture programs. Countries run agricultural agencies with programs for organic and sustainable transition, often including certification support, training, and sometimes input or equipment support. These are worth checking but competitive, and the timelines are long.

NGO and cooperative programs. Organizations working on farmer transition run training and sometimes support programs. These exist in most farming regions and are usually reached through a cooperative or a farmers association rather than individually.

Corporate sustainability programs. Companies with supply chains in specific crops sometimes fund farmer transition in those crops. Real, but crop-specific and relationship-driven: this reaches farms already organized, not individual applicants.

Two honest rules. First: a grant is a bonus, not a plan. The transition has to pay for itself through the input costs, or it will not survive the grant application's timeline. Second: never pay anyone to "help you access" a grant. The legitimate programs are free to apply for, and the people who know them are at your local agriculture office, not in a private consultant's invoice.

Dragonfruit: the cash-crop case

Dragonfruit is the worked example in this module because it shows how a cash crop and a regenerative system fit together. It is a vine, which means it grows on posts and occupies the vertical layer of the farm, the layer most smallholdings leave empty. It is a high-value crop on the market. And it produces for years from one planting, which means the establishment cost is paid once and the income runs.

On our farm it is the vine layer of the seven-layer planting: the canopy trees above it, the covers and roots below it, the animals in the system around it. The vine pays while the other layers build soil,

which is the whole argument for stacking: the cash crop funds the transition, and the transition makes the cash crop cheaper to run every year.

The same logic applies to whatever the high-value crop is in your region. The point is not dragonfruit specifically; it is that a cash crop in the vertical layer, on a farm whose input costs are falling, is what makes the transition's arithmetic work in the years before the soil's full answer arrives.

The numbers that decide it

Every farm's arithmetic is its own, and the honest tool is the cost notebook rather than anyone else's numbers. That said, the shape of the decision is consistent: the input costs (fertilizer, pesticide, fuel, irrigation) are the recurring bill the transition cuts first; the establishment costs (seed, posts for vines, the first compost pile) are small and one-time; and the income side (consistent harvest, premium buyers, direct sales) builds over the seasons. The farmer who writes their own numbers down for two seasons knows exactly where their farm stands, which is more than most farmers ever have.

The first season, in money

Because the first season is where farmers decide, it is worth spelling out what the money actually does in it. The establishment costs are small and one-time: a packet of cover crop seed, posts if a vine layer is going in, the first compost pile's labor. Against them, the recurring bill starts falling immediately: the first compost replaces a portion of the fertilizer purchase, the cover crop replaces a spray, the mulch replaces watering hours. Nothing in the first season is a big win; everything in it is a recurring cost getting smaller, which compounds.

The honest failure mode is the farmer who quits in the first season because the harvest did not jump. It was never going to; the harvest is the third or fourth thing to move, and the input costs are the first. A farmer who wrote their numbers down knows this, because the notebook shows the bill falling while the harvest holds steady, and that trade is the transition working exactly as designed.

The neighbors' economics

The transition spreads through a barangay the way it spread on our farm: not through advocacy, but through arithmetic that other farmers can check. The farm with the falling input bill and the field that held through the typhoon draws questions, and the questions are answered with the notebook, not with a speech. The Farmers Association is where those notebooks get compared, and it is the reason this method is taught publicly: one farm's numbers, visible, does more for the transition than any campaign.

How it connects to the rest of the method

Making it pay is module nine of ten, after the first-season plan and before measurement , because the economics only make sense once the system exists. The compost module is where the input costs fall, the cover crops are where the labor falls, and The Method course teaches the whole system.

Frequently asked questions

Are there grants for regenerative farming?

Yes, but no grant is guaranteed and they are country and region specific. The reliable sources are your local agriculture office, national agriculture programs, NGO and cooperative programs, and corporate sustainability programs in your crop. Start with your local office and your farmers association, and treat any grant as a bonus rather than a plan.

How does regenerative farming affect profit?

Input costs fall first, often in the first season, as compost and farm materials replace purchased fertilizer and pesticide. Labor drops as mulch and covers take over weeding and watering. Premium and direct sales build later, once the system produces consistently. The honest shape is: the budget improves before the harvest does.

Is dragonfruit profitable for a small farm?

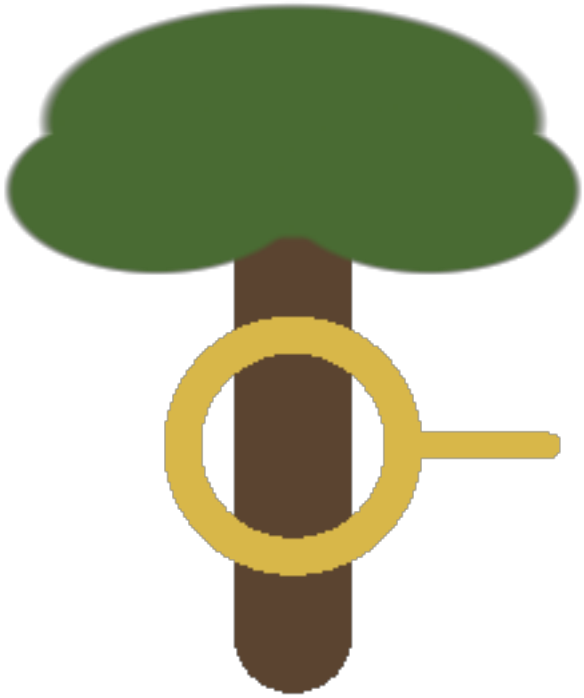
It is a high-value crop that produces for years from one planting and grows on the vertical layer most smallholdings leave empty, which is why it is the worked case here. Profitability depends on your market and your region, so check your local buyers and prices before planting. The lesson is the stacking, not the fruit.

How do I sell regenerative produce at a better price?

On verified practice, once the system produces consistently: direct to buyers, at markets where the growing story matters, or pooled through a cooperative that captures more of the margin. It depends on the measurement record, which is why the notebook comes before the marketing plan.

Chapter 10. Measure and Keep Going

Soil health indicators year over year, the farm notebook, cost tracking, and what to do when results do not show.



Short answer. You measure regenerative farming with a notebook and four tests, not a laboratory: a worm count, a jar test, a water infiltration test, and the smell and crumb of a handful of soil, written down with dates season by season. Add your input costs in the same notebook, and the trend, not a single number, tells you whether the farm is improving.

If you did not write it down, you cannot prove it happened.

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Why measurement is the module that keeps farmers in the transition

The regenerative transition has an uncomfortable first season and a rewarding second one, and the difference between the farmers who get to the second and the ones who quit is almost always measurement. A farmer who wrote down the worm count, the infiltration time, and the input costs in the first season has something to compare against in the second. A farmer who remembers a bad first season, and nothing else, quits on a feeling.

Measurement is also what makes the whole method honest. Claims about soil health are easy to make and hard to prove, and the reason this site can say "the soil improves" is that the

improvement is written down, season by season, in a notebook that costs nothing. A farm that measures can sell on its practice. A farm that guesses cannot.

The four tests, and how to run them

Each costs nothing, each takes minutes, and each is repeated in the same spot at the same time of season so the trend is real:

Run the worm count. Dig a spade of soil, 20 centimeters deep, and count the earthworms in it. Do it in the same field, the same month, every season. Zero is a warning, two or three per spade is workable, eight or more is a working soil. Earthworms are the visible face of the soil biology: where they thrive, the rest of the biology is thriving too.

Run the jar test. Fill a clear jar a third full with soil, top it with water, shake it hard, and leave it overnight. Sand settles first, then silt, then clay on top. Read the layers: they tell you how your soil drains, what it holds, and what your beds need. Do it once per field; the soil type does not change, but knowing it changes everything else you decide.

Run the water infiltration test. Push an open-ended can about ten centimeters into the soil, pour in a measured amount of water, and time how long it takes to disappear. Fast is good: it means water is moving down through open channels. Slow is compaction, and compaction is what the cover crops and worms fix. Time it in the same spot each season.

Read the smell and the crumb. Take a handful from spade depth. Healthy soil smells like a forest floor, dark and sweet. Degraded soil smells flat, sour, or like nothing at all. Squeeze it: healthy soil crumbles into crumbs when you open your hand; degraded soil shatters into dust or packs into a hard lump. This is the fastest test and the one to teach first.

Four lines in the notebook per field per season: worm count, infiltration time, smell, crumb. Written with the date and the weather. That record, kept for three years, is worth more to the farm than any single expensive analysis, because it is a trend rather than a snapshot.

The cost notebook: the other half of the record

Soil health is half the measurement. The other half is money, because the transition has to pay for itself or it does not survive the neighbors' opinion. In the same notebook:

Input costs per season, in your own currency. Fertilizer, pesticide, seed, fuel for the tiller. Written as spent, not estimated. The regenerative transition shows up here first: compost and biofertilizer replace purchases, the tiller idles, and the numbers fall.

Labor per task. Hours spent weeding, watering, and tilling. This is where the transition pays second: a mulched, covered farm needs fewer hours of weeding and watering than a bare one.

What you harvested, and where it went. The yield, honestly, season by season. The transition's first season is often flat or slightly down; the second and third are where the biology's work shows. Without the record, the flat first season is all a farmer remembers.

The cost notebook is also what makes the economics module real instead of theoretical. The farmer who can show their own numbers falling while their soil improves is the farmer who convinces the cooperative, the buyer, and the neighbor.

Honest timelines for what shows when

Within one season: the infiltration test and worm count start moving, because the cover crop is feeding the biology immediately. The smell follows. Weeds drop where the mulch went down.

Within a year: the crumb structure, visibly better water holding after rain, and the first real drop in input costs.

Year two to three: the topsoil darkens, the beds hold structure between seasons, and the yield stabilizes above the old average in good years and below it in fewer bad ones.

And the honest failure mode: if the numbers are flat after three honest seasons, something in the system is missing. Usually it is the cover layer, or the compost is not actually being made, or the soil was more degraded than the transition assumed. The record is what tells you which, and that is the point of keeping it.

What to do when results do not show

Three questions, in order. First: is the cover actually on the soil, all of it, all year? A partial cover gives partial results, and the gaps show up in the worm count first. Second: is the compost pile actually running, or did it stop in the wet season and never restart? Third: was the starting point more degraded than the transition assumed, in which case the timeline stretches but the direction does not change. The notebook answers all three, because it holds the record of what was actually done rather than what was intended.

The notebook on our farm

The notebook is worth describing concretely, because its plainness is the point. It is a school exercise book, one page per field per season, in the father's handwriting. Each page has four lines: worms per spade, how long the water took, what it smelled like, what the crumb did. Below them: what was spent on inputs, what was harvested, and what the weather did that season.

What that record changed: the barangay Farmers Association meetings went from opinion to evidence. Farmers comparing pages can see whose fields are improving and ask what they did differently, which is how the method actually spreads, farmer to farmer, page to page. And the farm's own decisions stopped being arguments with memory: the numbers settle whether the compost, the mulch, or the cover made the difference, because memory edits itself and paper does not.

The kit being designed for first-season farmers includes this notebook as one of its items, for exactly that reason: it is the cheapest thing on the farm and the one that makes every other change provable.

How it connects to the rest of the method

Measurement is module ten of ten, the last one, because it is the one that never stops. It uses the tests introduced in knowing your soil , it makes the economics real, and it is what lets a farm sell on verified practice, which the course teaches. The first-season plan is where the notebook starts.

Frequently asked questions

How do I measure soil health without a laboratory?

Four tests with household items: a worm count per spade, a jar test for soil type, a water infiltration test with an open can, and the smell and crumb of a handful from spade depth. Written down with dates in the same spot each season, they give a trend that is worth more than any single analysis.

How long before I see results in the soil?

The worm count and infiltration move within one season, the smell and crumb within a year, and the topsoil's visible darkening takes two to three years. The input costs fall first, often in the first year,

which is what keeps most farmers in the transition long enough for the soil to answer.

What if my numbers are not improving?

Three checks, in order: is the cover on all the soil all year, is the compost pile actually running, and was the starting point more degraded than assumed. The notebook answers all three, because it holds what was actually done rather than what was intended. Flat numbers after three honest seasons mean something in the system is missing.

Do I need to track money as well as soil?

Yes, in the same notebook. Input costs, labor hours, and harvests per season. The transition pays in the budget before it pays in the soil test, and the farmer who can show both moving is the one who convinces a cooperative, a buyer, or a neighbor.

Where to go from here

The full course, The Method, runs these ten chapters as modules with lessons, videos, and the season planner, at regenerativefarmingphilippines.com. The starter kit, being built now, covers what a first season needs in one box: hand tools, tropical-appropriate seed, compost starter materials, and the companion guide.

Join the Movement at regenerativefarmingphilippines.com for the next guides as they are finished, and the Founding Farmer rate when the course opens.

Regenerative Farming Philippines. Viking Farms PH, Calamba, Laguna.